Task 5: Mini AI Project – Rule-Based AI Chatbot

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Objective: To develop a basic chatbot that interacts with users and gives predefined responses using Python `if-elif-else` statements without Machine Learning.

1. Python Program (chatbot.py)

```
# Task 5: Rule-Based AI Chatbot

print("Welcome to Simple AI Chatbot!")
print("Type 'bye' to exit the chatbot.")

while True:
    user = input("\nYou: ").lower().strip()

    if user == "hello" or user == "hi":
        print("Bot: Hello! Welcome.")

    elif user == "how are you":
        print("Bot: I'm fine. Thank you!")

    elif user == "what is your name":
        print("Bot: I am a Simple AI Chatbot.")

    elif user == "what can you do":
        print("Bot: I can answer simple questions.")

    elif user == "thank you":
        print("Bot: You're welcome!")

    elif user == "bye":
        print("Bot: Goodbye! Have a nice day!")
        break

    else:
        print("Bot: Sorry, I don't understand that.")
```

3. Concepts Used

1. **input():** Accepts messages from the user in the terminal.
2. **print():** Displays chatbot responses back to the user.
3. **while loop:** Keeps the chatbot running continuously until an exit condition is met.
4. **if-elif-else:** Checks user messages against predefined conditions and selects appropriate responses.
5. **break:** Stops and exits the chatbot loop when the user types "bye".
6. **lower():** Converts user input into lowercase to ensure case-insensitive matching.
7. **strip():** Removes any leading or trailing extra spaces from user input.

4. Expected Interaction Flow Verification

```
Welcome to Simple AI Chatbot!
Type 'bye' to exit the chatbot.
You: hello
Bot: Hello! Welcome.
You: how are you
Bot: I'm fine. Thank you!
You: what is your name
Bot: I am a Simple AI Chatbot.
You: what can you do
Bot: I can answer simple questions.
You: thank you
Bot: You're welcome!
You: bye
Bot: Goodbye! Have a nice day
```

5. Conclusion

This mini project demonstrates how to create a simple Rule-Based AI Chatbot using Python. The chatbot interacts with users and provides predefined responses based on their input. It utilizes conditional statements and loops effectively without requiring complex Machine Learning models.

6. Implementation Screenshots

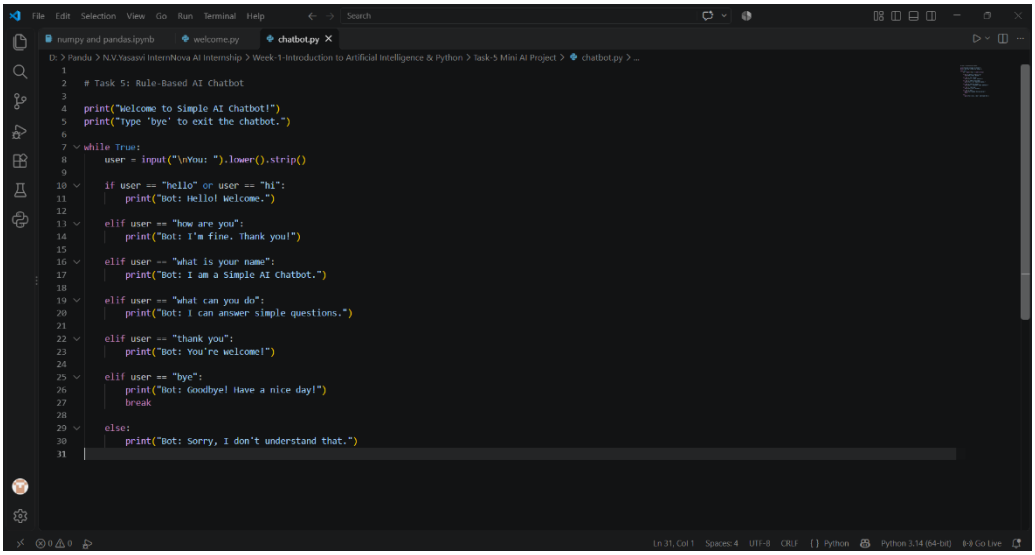


Figure 1: Python Source Code Implementation in VS Code (chatbot.py)



Figure 2: Terminal Conversation Output Between User and Rule-Based Chatbot